

Joint, Marginal and Conditional Distributions: a Summary

Tom Glinnan

September 2026

An introduction: what are we trying to describe?

These notes are a supplement to Lecture 2 of EC400. They assume that you have seen the lecture, but the aim is to spend a little more time on what the formulas mean and how to use them. In particular, the continuous case can feel like a large step up in difficulty. It helps to keep the discrete picture in mind: we will mostly be replacing sums with integrals, while keeping the underlying ideas the same.

The basics: what is a distribution?

A **random variable** attaches a number to an outcome. For example, if we select a student at random, X might record how many siblings they have. Before selecting the student, we do not know which value we will see. A **distribution** describes how the probability is spread across the possible values of X . It tells us which values, or ranges of values, are more or less likely. In lecture we described how to formally define the distribution of a random variable as a pushforward measure - but this is not necessary to understand all of the main ideas.

For a **discrete** random variable, we can list its possible values and the probability attached to each one. We write

$$p_X(x) = \mathbb{P}(X = x), \quad p_X(x) \geq 0, \quad \sum_x p_X(x) = 1.$$

These two properties are what make something a valid discrete distribution. The capital X denotes the random variable; the small x is a particular value it might take. The function p_X is called a probability mass function. The sum is over all possible values of X .

For a **continuous** random variable with a density, such as a waiting time, probability is represented by *area under a curve*:

$$\mathbb{P}(a < X < b) = \int_a^b f_X(x) \, dx, \quad f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) \, dx = 1.$$

Here $f_X(x)$ is a density height, and the latter two properties are analogously needed to have a well-defined continuous distribution. For example, if X is uniform on $(0, 2)$, the density is $1/2$ throughout this interval. The probability of falling between 0 and $1/2$ is the width times the height: $(1/2)(1/2) = 1/4$. An individual point has no width, so $\mathbb{P}(X = x) = 0$; as such, the best way to interpret a probability density is as a relative probability: if $f_X(x_1) = 2 \times f_X(x_2) > 0$ and the density is continuous at both points, you are approximately twice as likely to be in a sufficiently small neighbourhood of x_1 as in an equal-width neighbourhood of x_2 . This is true even though the probability of being at x_1 or x_2 themselves is literally 0. The cumulative distribution function, $F_X(x) = \mathbb{P}(X \leq x)$, records the probability accumulated up to x . It works in both cases: add up masses in the discrete case, or integrate the density in the continuous case.

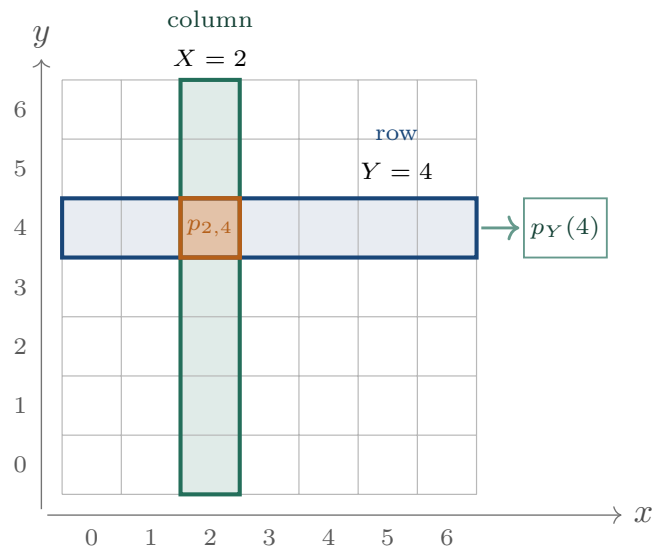
With two random variables, there are three natural ways we can slice things up: 1) **joint** distribution describes X and Y together, 2) **marginal** distribution describes one of them on its own and 3) the **conditional** distribution describes one of them when we know something about the other.

Discrete distributions: the table view

Suppose, as in lecture, that X and Y each take values $0, 1, \dots, 6$. The **joint probability mass function** assigns a probability to each possible pair:

$$p_{x,y} = \mathbb{P}(X = x, Y = y).$$

The comma means “and”: both statements must hold. The **joint distribution** is the full collection of these probabilities - in other words, the set of $7^2 = 49$ numbers that $p_{x,y}$ can take, as x and y each range from 0 to 6. Every entry is non-negative, and $\sum_x \sum_y p_{x,y} = 1$. The grid itself - in other words, the 49 different co-ordinates (i, j) for $i = 0, \dots, 6$ and $j = 0, \dots, 6$ is the set of outcomes $\omega \in \Omega$, from the formal definition of the random variable.



The lecture’s picture gives us three useful ways to read the table. X runs along the horizontal axis, and Y up the vertical axis.

The orange cell describes $X = 2$ and $Y = 4$ together. Its probability is $p_{2,4}$.

The blue row contains every way that $Y = 4$ can happen, across all values of X .

The green column contains every possibility that remains once we know $X = 2$.

Marginal distributions: add over the other variable

To find $\mathbb{P}(Y = 4)$, add the probabilities in the blue row. This makes sense because the cells are mutually exclusive: we cannot have $X = 0$ and $X = 1$ at the same time. More generally,

$$p_Y(y) = \sum_x p_{x,y}, \quad p_X(x) = \sum_y p_{x,y}.$$

These are the **marginal distributions**. They describe each variable on its own, after adding over all possibilities for the other variable. In this picture, $p_Y(y)$ is a row total and $p_X(x)$ is a column total.

Conditional distributions: keep a column, then rescale

Now suppose we are told that $X = 2$. As such, there are now only seven outcomes which could possibly occur: the seven cells which are within the green column. However, the numbers inside the column themselves do not make up a valid distribution, as they do not sum to one. Instead, they add up to $p_X(2)$, which is generally less than one. We therefore divide every entry by the column total:

$$p_{Y|X}(y | 2) = \mathbb{P}(Y = y | X = 2) = \frac{p_{2,y}}{p_X(2)} = \frac{p_{2,y}}{\sum_{r=0}^6 p_{2,r}}, \quad p_X(2) > 0.$$

The symbol $|$ means “given”. Note that r in the denominator is just a summation index: we add over the whole column. Dividing by its total ensures that $\sum_y p_{Y|X}(y | 2) = p_X(2)/p_X(2) = 1$, and so $Y | X$ now is a valid distribution, which has the two properties that a distribution needs to have (see above). Put another way, conditioning means restricting attention and then rescaling. Marginalising means adding over possibilities. Keeping these two operations separate in your mind is much more useful than simply learning the formulas.

A small worked example

Consider the following joint distribution. The six interior entries are joint probabilities; the margins contain their totals.

$Y \backslash X$	0	1	2	$p_Y(y)$
1	0.10	0.20	0.20	0.50
0	0.20	0.20	0.10	0.50
$p_X(x)$	0.30	0.40	0.30	1.00

For example, $\mathbb{P}(X = 2, Y = 1) = 0.20$. Adding along the $Y = 1$ row gives $\mathbb{P}(Y = 1) = 0.10 + 0.20 + 0.20 = 0.50$. If instead we know $X = 2$, we use only the $X = 2$ column and divide by its total, 0.30:

$$\mathbb{P}(Y = 0 | X = 2) = \frac{0.10}{0.30} = \frac{1}{3}, \quad \mathbb{P}(Y = 1 | X = 2) = \frac{0.20}{0.30} = \frac{2}{3}.$$

These two numbers form the entire conditional distribution of Y given $X = 2$. They add to one, as they should. In a Bayesian interpretation of probability, we can think of the marginal distribution as our beliefs ‘before’ seeing the other variable, and the conditional distribution as our beliefs ‘afterwards’ - for example, we could say that learning $X = 2$ changes the probability of $Y = 1$ from $1/2$ to $2/3$. This perspective can be very helpful when thinking about how to interpret the Gauss Markov assumptions (ie the assumptions under which OLS works well), as these are often written in this language.

Conditional probability: why divide by the denominator?

The same reasoning applies to any two events A and B . An event is a set of outcomes; $A \cap B$ means that both events occur, and B^c means that B does not occur. Once we know B has happened, we restrict attention to B . The part of this set in which A also happens is $A \cap B$. Its share of the probability in B is

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad \mathbb{P}(B) > 0.$$

For equally likely outcomes, this is simply the number of outcomes in $A \cap B$ divided by the number in B . With unequal probabilities, the formula does the same calculation using probability mass. The key is just to see that this is the analogous thing to what it means to have a conditional distribution. Two more things to note: firstly, assuming also that $P(A) > 0$, note that we can write $P(A \cap B)$ as both $P(A | B) \times P(B)$ and $P(B | A) \times P(A)$; as we showed in lecture, therefore writing $P(A | B) \times P(B) = P(B | A) \times P(A)$ and dividing by $P(B)$ gives **Bayes' rule**

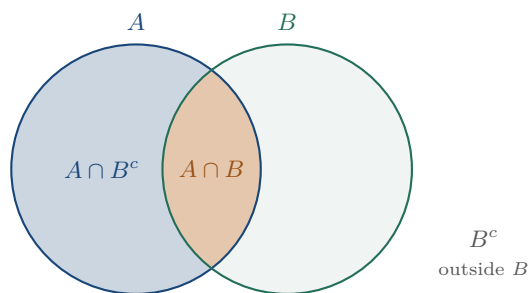
$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

Secondly, recall that independence is a setting where learning B does not change your opinion about A . As such, it's defined as when $P(A | B) = P(A)$, so multiplying both sides by $P(B)$ gives $P(A \cap B) = P(A) \times P(B)$.

Splitting an event: the law of total probability

Every outcome in A either belongs to B or to B^c . These possibilities cover all of A and cannot overlap. Therefore,

$$\begin{aligned} A &= (A \cap B) \cup (A \cap B^c), \\ \mathbb{P}(A) &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c). \end{aligned}$$



The blue and orange regions together form all of A . Their probabilities add because the two pieces do not overlap.

Rearranging the definition of conditional probability gives $\mathbb{P}(A \cap B) = \mathbb{P}(A | B)\mathbb{P}(B)$. Applying this to both pieces yields

$$\mathbb{P}(A) = \mathbb{P}(A | B)\mathbb{P}(B) + \mathbb{P}(A | B^c)\mathbb{P}(B^c),$$

when both conditioning events have positive probability. This is the **law of total probability**. We find the probability within each group and weight it by the probability of belonging to that group. For a finite or countable collection of disjoint groups $B_1, B_2, \dots$ covering all outcomes, the same argument gives

$$\mathbb{P}(A) = \sum_j \mathbb{P}(A | B_j)\mathbb{P}(B_j).$$

Groups with probability zero can be omitted. The marginal row total above is an example: the groups are the different possible values of X .

A worked example: a signal about a borrower

Take the example from lecture. Let H be the event that a borrower will default, and S the event that we observe a risk signal. Suppose

$$\mathbb{P}(H) = 0.10, \quad \mathbb{P}(S \mid H) = 0.80, \quad \mathbb{P}(S \mid H^c) = 0.20.$$

We observe the signal and want $\mathbb{P}(H \mid S)$. Before calculating anything, notice the direction of the conditioning. We are given the probability of a signal *among borrowers who will default*; we want the probability of default *among borrowers who produce a signal*. There is no reason for these numbers to be the same.

Step 1: find the two ways a signal can occur. A signal can come from a borrower who will default, or from one who will not. First, $\mathbb{P}(H^c) = 1 - 0.10 = 0.90$. Using the multiplication rule,

$$\begin{aligned}\mathbb{P}(S \cap H) &= \mathbb{P}(S \mid H)\mathbb{P}(H) = 0.80(0.10) = 0.08, \\ \mathbb{P}(S \cap H^c) &= \mathbb{P}(S \mid H^c)\mathbb{P}(H^c) = 0.20(0.90) = 0.18.\end{aligned}$$

Step 2: add the pieces to find the denominator. Apply the decomposition on the previous page with $A = S$ and $B = H$:

$$\mathbb{P}(S) = \mathbb{P}(S \cap H) + \mathbb{P}(S \cap H^c) = 0.08 + 0.18 = 0.26.$$

This is an application of the law of total probability. The 0.80 and 0.20 are weighted by the sizes of their respective groups.

Step 3: restrict attention to borrowers with a signal. Of the probability mass 0.26 in S , the part corresponding to default is 0.08. Hence

$$\mathbb{P}(H \mid S) = \frac{\mathbb{P}(H \cap S)}{\mathbb{P}(S)} = \frac{0.08}{0.26} = \frac{4}{13} \simeq 0.308.$$

One way to make this concrete is to imagine 1,000 borrowers with exactly these proportions. Of the 100 who default, 80 produce a signal; of the 900 who do not, 180 produce a signal. There are 260 signals altogether, and 80 of them come from borrowers who default. We get the same answer, $80/260$. The signal raises the default probability from 10% to about 31%.

Continuous distributions: from cells to a surface

For two jointly continuous random variables, a joint density $f_{X,Y}(x,y)$ describes how probability is spread over pairs of values. The height of the surface is a density; probability is the *volume underneath it* over a region. For example,

$$\mathbb{P}(a < X < b, c < Y < d) = \int_a^b \int_c^d f_{X,Y}(x,y) \, dy \, dx.$$

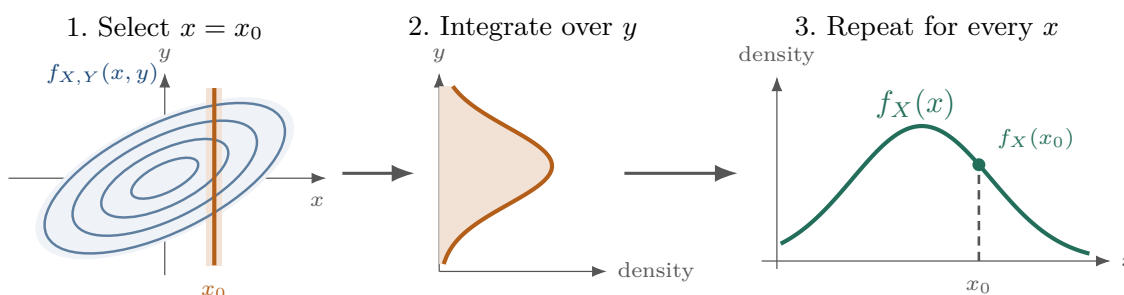
The density is non-negative and integrates to one over the whole plane. This is the continuous counterpart of requiring all the cells in the joint table to add to one.

Marginal densities: integrate over the other variable

To describe X on its own, we add up the density over all possible values of Y . In the continuous case, this means integrating:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) \, dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) \, dx.$$

Keep track of which variable you want to be left with. To find $f_X(x)$, hold x fixed and integrate out y . The result can still depend on x ¹.



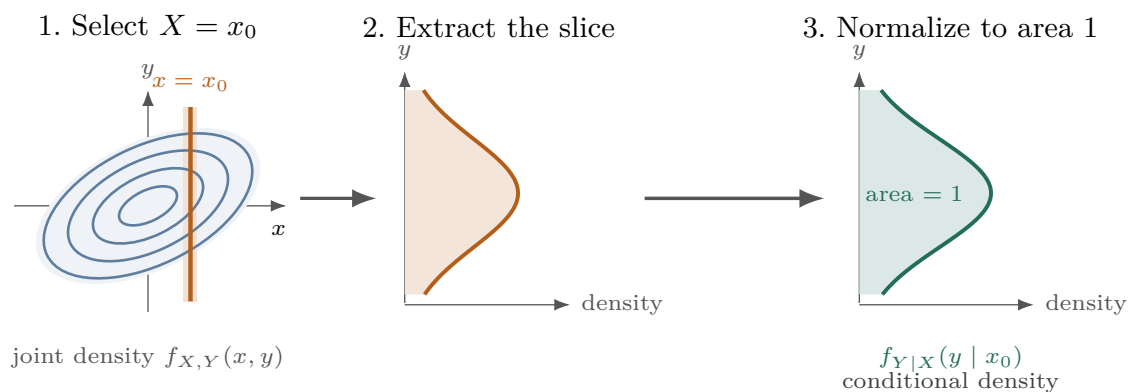
The lecture's diagram views the joint density from above (just like we did with multivariate normals), with the orange line selecting $x = x_0$. The middle panel shows the density along that slice, drawn sideways. Its area supplies one height, $f_X(x_0)$, of the marginal curve on the right. Repeat for every x to obtain the whole function. Note that $f_X(x_0)$ is ultimately a probability density, and so should be interpreted in the careful way we've seen in lecture (i.e. as a relative probability of being close to x_0 , rather than probability of being x_0 itself).

¹Independence instead means that the conditional distribution of Y given $X = x$ does not vary with x .

Conditional densities: rescale a slice to have area one

To condition on $X = x_0$, keep the same slice, but now use it to describe how Y varies at this particular value of X . Its area is $f_X(x_0)$, so divide by this number:

$$f_{Y|X}(y | x_0) = \frac{f_{X,Y}(x_0, y)}{f_X(x_0)}, \quad f_X(x_0) > 0.$$



Just like in the discrete case, the main idea is that dividing by the marginal density makes the conditional density integrate to 1, and so it has the properties that a probability density needs to have. As you might imagine, conditional probabilities are areas under this new density:

$$\mathbb{P}(a < Y < b | X = x_0) = \int_a^b f_{Y|X}(y | x_0) \, dy,$$

exactly by analogy to the discrete case.

A continuous worked example: start by drawing the support

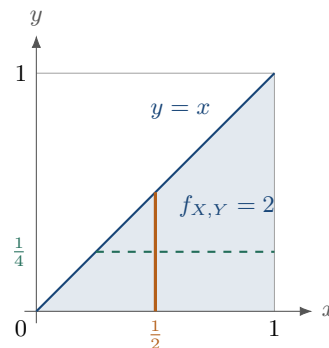
Suppose the joint density is

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The **support** is the set of values of (x, y) for which $f_{X,Y}(x, y) > 0$. Here it is the triangle below $y = x$ in the unit square. The density is constant over this triangle. Its total probability is height times area, $2(1/2) = 1$. Equivalently,

$$\int_0^1 \int_0^x 2 \, dy \, dx = \int_0^1 2x \, dx = 1.$$

I would always recommend sketching the support before integrating. It tells you which values to include and, therefore, what your integration limits should be.



Step 1: find the marginal densities. For a fixed x between zero and one, the vertical slice runs from $y = 0$ to $y = x$. For a fixed y between zero and one, the horizontal slice runs from $x = y$ to $x = 1$. Thus

$$\begin{aligned} f_X(x) &= \int_0^x 2 \, dy = 2x, & 0 < x < 1, \\ f_Y(y) &= \int_y^1 2 \, dx = 2(1 - y), & 0 < y < 1. \end{aligned}$$

Both densities are zero outside these intervals. Larger x gives a longer vertical slice, so $f_X(x)$ rises with x ; larger y gives a shorter horizontal slice, so $f_Y(y)$ falls with y . This is a useful check on the algebra. Each marginal also integrates to one.

Step 2: divide by the relevant marginal. To condition on $X = x$, divide by $f_X(x)$:

$$f_{Y|X}(y | x) = \frac{2}{2x} = \frac{1}{x}, \quad 0 < y < x, \quad 0 < x < 1,$$

and the conditional density is zero for other y . Hence $Y | X = x$ is uniform on $(0, x)$. Its area is $\int_0^x (1/x) \, dy = 1$. Knowing x tells us the upper limit of the possible values of Y .

Step 3: use the density to calculate a probability. At $X = 1/2$, the conditional density is 2 on $(0, 1/2)$. Therefore,

$$\mathbb{P}(Y < \tfrac{1}{4} | X = \tfrac{1}{2}) = \int_0^{1/4} 2 \, dy = [2y]_0^{1/4} = \frac{1}{2}.$$

The threshold cuts the possible interval for Y in half. Without conditioning, we instead use the marginal density:

$$\mathbb{P}(Y < \tfrac{1}{4}) = \int_0^{1/4} 2(1 - y) \, dy = [2y - y^2]_0^{1/4} = \frac{7}{16}.$$

I'd encourage you to practise a few of these questions, and you'll see that it will start to make much more sense. Please do ask in seminars if anything is unclear!